

Survey Design Seminar

Survey Methods 670

Winter 2026

Week 2: Types of consulting and problem-solving arrangements

January 15, 2026

Outline of Today's Class

- Discussion of next week's client presentations
- Availability of extra readings
- Introduction to consulting
 - Types of consulting and problem-solving arrangements
 - Work initiation
 - Responding to work help requests
 - Sources of requests for help:
 - 1- Funding work
 - 2- Types of work
 - Statistical consulting for the media
- Uniqueness vs. generality

Next week's client presentations

- General organization

Client presentation followed by student Q&A

Advance work/potential follow up

Do you know what the main concepts are?

Do you have a general idea of a design?

Can you think of possible products?

How does the client presentation affect these preliminary views?

- Organization of teams will follow preference choices
- Preparation of follow up emails

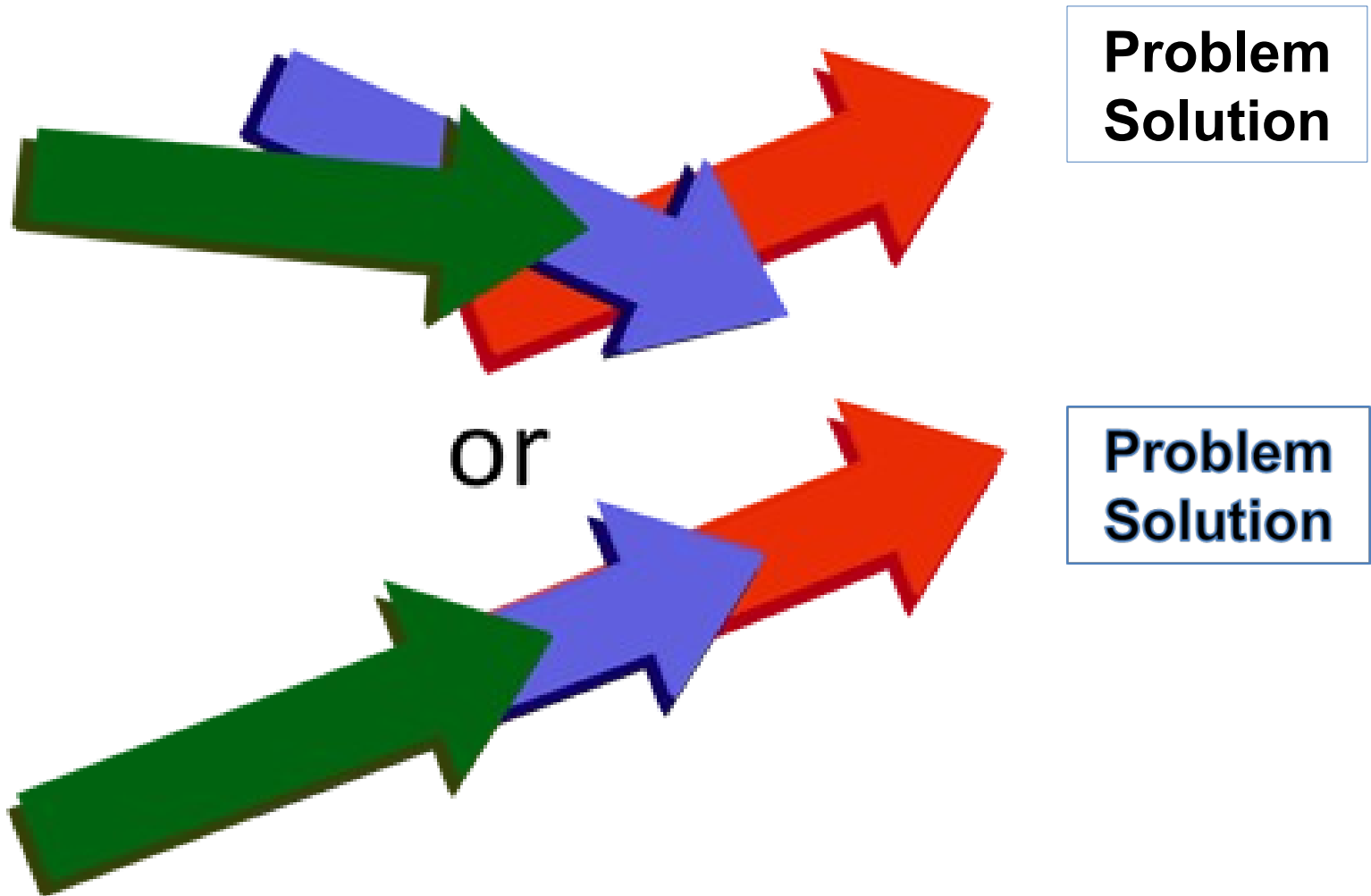
Expected Order of Client Presentations

HRS	Brady West	9:30-9:50
SurveyMonkey	Wendy Smith	9:50-10:10
African Surveys	Justine Davis	10:10-10:30
ASHA	Shari Jurik	10:30-10:50
PSID	Tom Crossley	10:50-11:10
A2 Rotary Club	Rona Hu	11:10-11:30

Introduction to Consulting

- Types of consulting and problem-solving arrangements
 - University setting vs. others
 - External vs. internal
 - Design consulting, data collection consulting, and statistical analysis consulting
- Definition of an expert
- Navigating across different worlds

Why Engage a Consultant?



6 Steps in Successful Consulting

- **Establish rapport**
- **Identify the problem, agreement with client**
- **Set goals for team and members**
- **Agree on responsibilities within the team**
- **Preview the results**
- **Present the agreed upon product(s)**

Different Consulting Arrangements

Types of consulting

Academic

To and in industry

Statistics in public media

Statistical Consulting in Academia

From a faculty member's perspective

Teaching: undergraduates and grad students

Research: producing new knowledge

**Service: administrative support for consulting,
effective public communication of results**

Statistical Consulting in Academia: At Michigan – A Time of Change



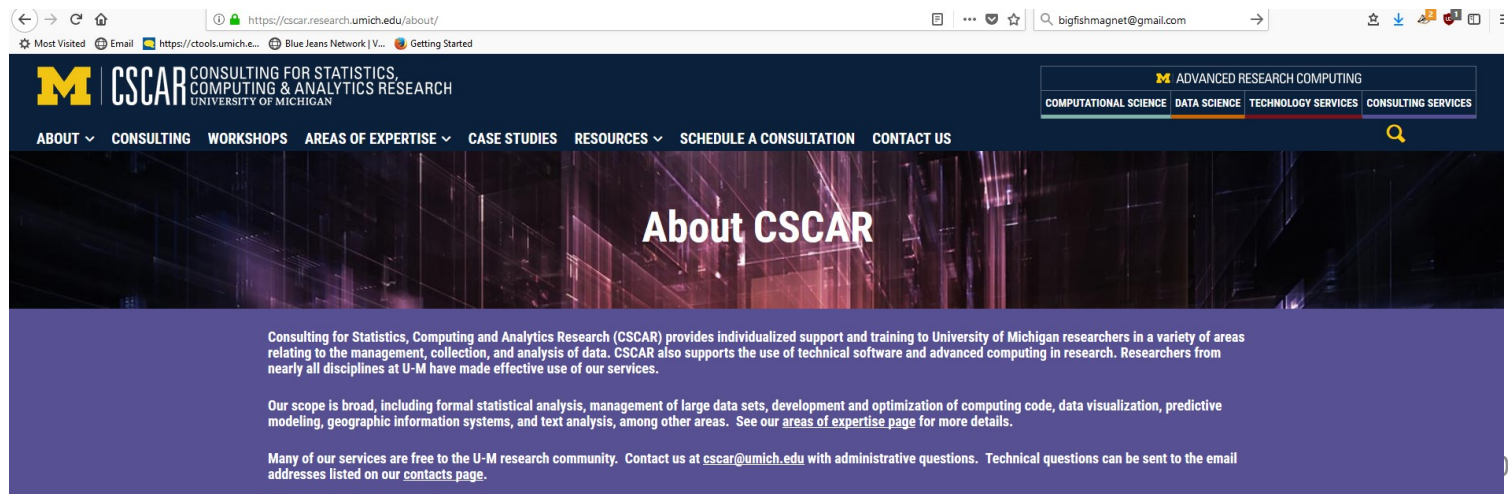
Schedule a Consultation

Please use our [Request Assistance form](#) or email deskpeople@umich.edu to set up an appointment.

You will be scheduled to meet with a CSCAR consultant. Please share a brief description of your research aims and what software you will be using; let us know what you wish to accomplish during the consulting appointment.

CSCAR AREAS OF EXPERTISE

UM researchers are encouraged to make use of CSCAR services as needed; We typically limit each researcher to one appointment per week.



Statistical Consulting in Academia: At Michigan – A Time of Change

CSCAR Update

Research universities do amazing things. They save lives, create opportunity, and improve quality of life for people all over our state, across the nation, and around the world. The Office of the Vice President for Research provides a wide range of essential services to principal investigators of externally funded research grants to help the University of Michigan do groundbreaking, life-saving work in so many areas.

As you may also know, research universities like ours operate in an increasingly challenging environment. Terminations, investigations, and new regulations are just some of the new realities that we must manage. In light of funding challenges associated with these changes, the research office had to review its entire portfolio of services to clarify the things for which we are primarily responsible at this moment. Part of this review examined benchmarking statistical and data services across campus and listening to members of our community who offer or receive these services in different ways. As a result of this review, the office made the decision to wind down its financial support of CSCAR over the coming year.

In past decades, CSCAR was the only campus source for statistical consulting. That is no longer the case. So, since earlier this year, we have been coordinating with campus partners and others to identify ways for participating units to share the costs of CSCAR or to take it over. We have not been successful. While we are able to offer continuing support to researchers currently working with CSCAR on grants for the duration of those grants, we have turned our focus to finding alternative options for others to find statistical and coding advice. There are many sources available on campus and online. In the coming weeks, we will roll out detailed information on how to find them. We are grateful to everyone involved in this effort.

Making decisions like this is difficult. At the same time, these decisions are essential for the research office to manage challenges to funding environments that have arisen in recent months — and to focus on areas where OVPR has the primary responsibility to protect and advance the university's research mission.

[Statistical Consulting Options – Updated as New Resources Emerge](#)

Statistical Consulting in Academia: At Michigan – A Time of Change

[Statistics Online Computational Resource \(SOCR\)](#)

Contact: Ivo Dinov

Email: dinov@umich.edu

Summary of Services:

The Statistics Online Computational Resource (SOCR) provides a wide range of statistical, informatics, computing, analytic and AI consulting services. Consultants include SOCR faculty, research staff and students. Consulting services are offered to academic trainees, scholars, centers and institutes, state and government organizations, industry partners, and non-profit entities. We have mechanisms to provide percent-effort commitments of faculty, research staff, as well as students (GSRA). Typically, these involve longer agreements, but we may provide some shorter and pro bono consults, as necessary.

Example of SOCR consulting services include:

- Research study design
- Biostatistical (e.g., statistical-power, parametric/non-parametric tests, statistical analyses)
- Data science (e.g., exploratory data analytics, visualization, ML/AI, statistical forecasting, computational prediction, math modeling)
- Data management
- Foundational and generative AI modeling (GAIM)
- Informatics (e.g., biomedical, health, clinical, analytics)
- Other (see the SOCR Research Projects)

Statistical Consulting in Academia: At Michigan – A Time of Change

Institute for Healthcare Policy and Innovation – Data & Methods Hub

Contact: Patrick Brady
Email: pbrady@umich.edu

Summary of Services:

The IHPI Data and Methods Hub provides access to data and analytic resources primarily aimed at health services researchers. Our data resources include large insurance claims data and electronic health record datasets covering large portions of the US population. Our services are available to University of Michigan faculty who are also members of IHPI. For more information, please visit <https://ihpi.umich.edu/data>

ISR Survey Research Center

Contact: Pamela Davis-Kean
Email: pdakean@umich.edu

Summary of Services:

Help with designing surveys:

- <https://src.isr.umich.edu/services/survey-research-operations/> (Stephanie Chardoul)
- Statistical Consultation: <https://pdhp.isr.umich.edu/resources/> (Brady West)
- This is another link for stats help resources at ISR: <https://datascience.isr.umich.edu/events/coderspaces/>

Statistical Consulting in Academia: University of Maryland

Citation Tools

GIS and Spatial Data Center

Statistical Consulting

Graduate School Writing Center

Statistical Consulting

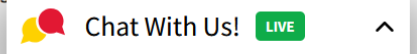
The **GIS and Data Service Center** offers fundamental statistical consulting service through lecture series, one-on-one consultation, data help desk.

What we do

- ▶ Answer your statistics and analysis questions through:
 - Virtual consultation using Zoom
 - In-person consultation at McKeldin library
 - **Data Help Desk** (Virtual), every Thursday (noon-2pm)
 - Request consult: **Intake Form**
- ▶ We currently specialize in **SAS**, **R**, and **Python** among others
- ▶ We do NOT tutor on statistical classes or homework.

Got a statistical question?

Research is increasingly data-driven, and we are available to answer your statistics and analysis questions. If you have a question, please fill out this **service intake form**.



<https://www.lib.umd.edu/research/services/statistical-consulting>

Statistical Consulting in Academia: University of Maryland

Home

Staff

Expertise

Services

Fee Structure

Outreach

Mission Statement

Our mission is to advance quantitative research at the University of Maryland and throughout the broader Maryland community by providing high-quality statistical consulting and training. We support researchers by offering:

- Free 1.5 hour project consulting session for campus students working on research projects
- Paid consulting at an affordable rate for on-campus and sister-campus student and faculty research projects
- Paid consulting on statistical projects for government-agency, nonprofit and commercial clients
- Collaborative consulting toward the writing of research contracts with statistical participation in study design and/or analysis
- Training of Statistical Students in the Practice of Statistical Consulting
- Outreach on campus in the form of tutorial lectures and short-courses on applied statistics

Who We Serve

- Students and Researchers with Quantitative Research Projects on the UMD Campus
- Off-campus clients needing statistical expertise
- UMD students and researchers through lectures and short-courses

What We Provide

- Advice on design of data collection
- Guidance on statistical modeling and data analysis
- Help with workflow, data structures, models, and interpretation

Different Consulting Roles

Helper: Low level of personal involvement

Leader: Make sense out of the data

Data Blessor: Is my design/analysis appropriate?

Collaborator: Pooling talents and expertise as partners

Teacher: Developing the client's skills

Private Sector Statistical Consulting

Our Leadership & Staff

Executive Team

[Michael Dimock](#), President

[Jess Awtry](#), Vice President, Digital and Communications Strategy

[Claudia Deane](#), Executive Vice President

[Courtney Kennedy](#), Vice President, Methods and Innovation

[Neha Sahgal](#), Vice President, Research

Managing Directors

[Monica Anderson](#), Director, Internet and Technology Research

[Alan Cooperman](#), Director, Religion Research

Alycia Kantor, Director, Legal Affairs

[Jocelyn Kiley](#), Director, Politics Research

[Mark Hugo Lopez](#), Director, Race and Ethnicity Research

[Katerina Eva Matsa](#), Director, News and Information Research

Christie Moon, Director, Human Resources

[Kim Parker](#), Director, Social Trends Research

[Stacy Rosenberg](#), Director, Digital Strategy

[Aaron Smith](#), Director, Data Labs

[Blythe Thomas](#), Director, Communications Strategy

[Richard Wike](#), Director, Global Attitudes Research

Nikolas Wissmann, Director of Financial Administration

Private Sector Statistical Consulting

<https://www.abtglobal.com/services>



Digital & Data Services



**Research, Monitoring &
Evaluation**



**Technical Assistance &
Implementation**



Agile IT Solutions

What do staff statisticians do?

- Attend staff meetings with active engagement
- Provide education and training
- Be available for informal consultation
- Review and help to write proposals
- Consult as a project collaborator
- Help to develop protocols, forms, and hypotheses
- Keep up with new developments
- Conduct independent methodological research
- Write up results for presentation or publication
- Serve in professional associations

Academic and Industry Consulting Opportunities (From the Consultant's Perspective)

NON-COMPETITIVE

Reputation
Prior relationships
Niche marketing
Prepare proposals



AWARD

COMPETITIVE

Locate opportunities
Prepare proposals
Bidders' meetings
Client presentations



AWARD

Stages in the Consulting Process



Locating New Opportunities

- Federal government small business resources:

<https://www.sba.gov/federal-contracting>

Contracting Resources for Small Businesses:

- FedBizOpps: Federal Business Opportunities

<https://sam.gov/opportunities>

Request for information (RFI) example:

<https://sam.gov/opp/f71a6ca63080457b953056440f6b42f9/view>

- Pre-solicitation notice example:

<https://sam.gov/workspace/contract/opp/4d5849ae92224c2c96ee196428f8e2a9/view>

Locating New Opportunities

- The Institute for Public Procurement

<http://www.nigp.org/>

- State procurement websites:

<http://www.nigp.org/home/find-procurement-resources/directories/state-province-procurement-websites>

<https://www.michigan.gov/dtmb>

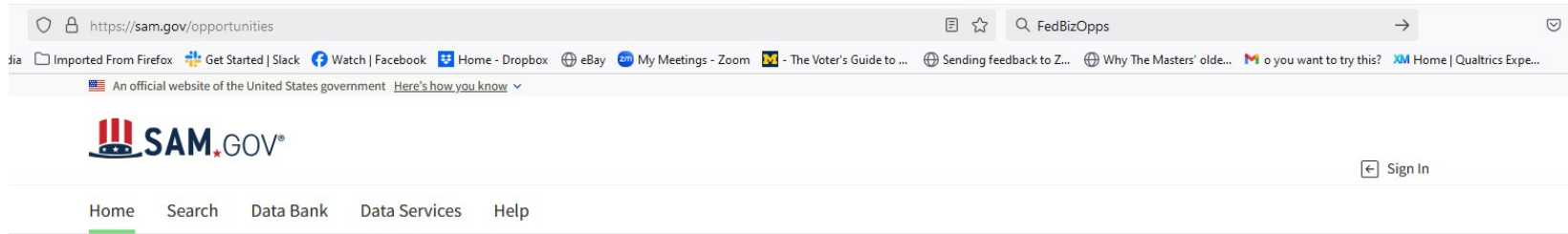
- Statcom at UofM

<https://sph.umich.edu/biostat/statcom/>

- UM Library

<https://guides.lib.umich.edu/c.php?g=283405&p=1888659>

Locating New Opportunities



Contract Opportunities

Contract opportunities are procurement notices from federal contracting offices. Anyone interested in doing business with the government can use this system to search opportunities. Opportunities include pre-solicitation notices, solicitation notices, award notices, and sole source notices.

If you are looking for grant opportunities, visit our partner site at grants.gov.

Federal Users

Manage contract opportunities as a contracting officer, contracting specialist, or administrator.

[Sign in to get started](#)

[Go to Contract Opportunities Federal home](#)

Search Contract Opportunities

[Advanced Search](#)

e.g. W91QVN-17-R-0088, light bulbs



☒ Show active only

[+ Get started searching contract opportunities](#)

[+ Use contract opportunities advanced search](#)

Do I need a user account?

Anyone may search contract opportunities without an account. A user account lets you save searches, follow changes to opportunities, and join interested vendor lists.

Help

[? Frequently Asked Questions](#)

[📖 Glossary Terms](#)

Video: Advanced Search of Contract Opportunities

Locating New Opportunities

<http://www.grants.gov/>

Grants.gov

[Home](#) [Search](#) [About](#) [Community](#) [Sign in](#)

Search funding opportunities

[Return to Grants.gov](#)

Tip: Use a minus sign to exclude words or phrases, like "-research"

Search

Filters

Copy this search query

☒ Must include all words (ex. transportation AND safety)

☐ May include any words (ex. transportation OR safety)

Forecasted

Open

1,589 Opportunities

1 2 3 4 5 ... 64 Next

Export results

Sort by Most relevant (Default)

Close date	Status	Title	Agency	Award min	Award max
Jan 13, 2026	Open	★ RESEARCH INITIATIVES AT THE NAVAL POST GRADUATE SCHOOL Number: N00244-22-S-NPS-F001	Naval Supply Systems Command Posted date: Jan 13, 2022 Expected awards: 40		
Jan 13, 2026	Open	★ ROSES 2025: A.2 Earth Venture: TROPICS and PREFIRE Science and Applications Teams Number: NNH25ZDA001N-EVPT	NASA Headquarters Posted date: Jul 11, 2025 Expected awards: --	\$--	\$--
Jan 13, 2026	Open	★ Oyster Pads and Wave Trip at Long Bird Island, Houston Ship Channel Number: W9126G262S0I7308	Fort Worth District Posted date: Dec 4, 2025 Expected awards: 1	\$100,000	\$533,000
Jan 13, 2026	Open	★ Pathways to Enable Open-Source Ecosystems Number: 24-606	U.S. National Science Foundation Posted date: Sep 14, 2024 Expected awards: 50	\$300,000	\$1,500,000
		Cooperative Agreement for Affiliated	Geological Survey		

Statistical Consulting for the Media

Polls v. results

Inbox x



Fessenden, Ford <fordfess@nytimes.com>

Jan 12 (6 days ago)



to me

We are trying to marshal some precinct level results from Michigan and other states to possibly explain better or at least more deeply the voting pattern that won the presidency for Donald Trump, and that confounded the conventional wisdom about polling.

Interested in something you said to the freep soon after the election about the secret sauce of deciding who is going to vote.

do you think now that this is the reason for the failure of polls to predict? did pollsters adequately understand the feelings of a certain group of voters, but fail to predict their turnout? is it that simple?

and here's another question: would precinct level results help to answer that question?

Michael Traugott <mtrau@umich.edu>

Jan 12 (6 days ago)



to Ford

Ford - Thanks for your email. There are three major factors that can affect the accuracy of a pre-election poll estimate of an election outcome: the quality of the sample, the weighting algorithm, and the likely voter model. If you look at the Bloomberg Politics Poll Decoder (<https://www.bloomberg.com/politics/graphics/2016-poll-decoder/>), you will get an idea of the variance in the samples (allowing for weighting) in the support for Clinton and Trump among various groups that supported them as well as those that weren't leaning either way. For example, we know a lot about the gender gap, and it should have favored Clinton. Across a group of late polls, her average advantage among women was about 12.6 percentage points, but in one sample it was 17 pp and in the lowest it was 6 pp. A second issue is weighting, and the standard approach is to weight to know population characteristics for which there is an external source like the Census. These factors include age, race, sex, and region. But there are other factors that some pollsters use like past voting behavior or party identification. There are no equivalent external distributions on which to base such weights, and issues can arise (see the example of the USC Dornsife panel weighting issues).

The third issue is that voting is a two-step process in the US: you have to get registered and then get a ballot. Turnout in the US is slightly greater than 50% in the US among the voting eligible population (around 59% in 2016 according to McDonald - <http://www.electproject.org/2016g>). So here is where the secret source comes in, because there is no standard model for estimating likelihood to vote. Usually the models involve a variety of factors: self-reported registration status, interest in the campaign (a survey measure), past voting behavior, and self-described likelihood of voting (subject to strong social desirability influences). What is the reference point for pollsters using LV models? Typically it is the last or recent elections (I just decided to add some slides from various post-election talks I gave; I think they are probably self-explanatory in the context of this email). Obama's elections turned out not to be very good predictors of voting because the Democrats lost votes and vote share in 2012 and 2016. The Republicans may have turned out people who didn't vote in 2008 or 2012 but did in 2004. We also saw for the first time a different turnout pattern in rural areas (higher) than in urban areas (lower) that caught many political observers by surprise after the fact. I have also come to believe since 2012 that the candidates' emphasis on the battleground states and the differentially higher turnout there compared to the rest of the country may mean that they should be treated as a separate stratum with a different likely voter model than the rest of the country.

Having said all of this, the national estimates of the popular vote - after all the votes were counted - were very accurate. The problem was lower quality estimation in the states. And the aggregators and modelers who used state level polls uniformly got in trouble. Beyond this, there is no accounting for the Clinton campaign not advertising in Wisconsin or visiting there; a similar question about Michigan.

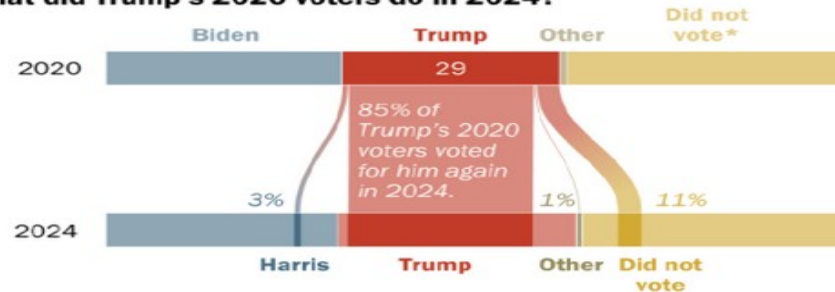
Now back to your original question about the use of precincts. They are by definition aggregates, so you can only make inferences about voting behavior in them when they are homogeneous. Comparing them over time, you could detect some patterns by size of place or perhaps race/ethnicity. But they won't tell you anything about voting patterns by gender or age, for example, nor about people who held different attitudes about government or specific policies. Only survey data at the individual level can enlighten you there.

Let me know if this helps or if you have additional questions. Mike

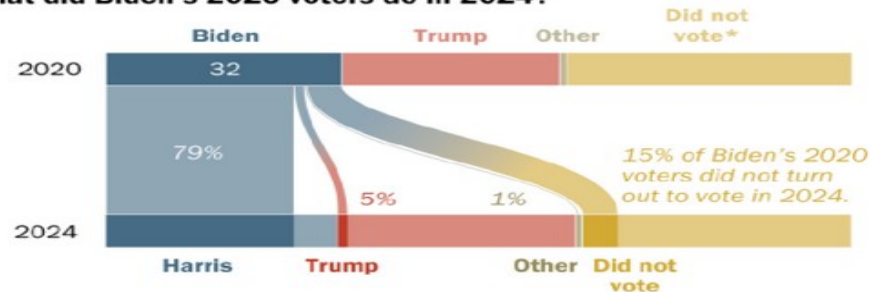
Statistical Consulting for the Media: Visualizations at Pew Research

The flow of voters and nonvoters from 2020 to 2024

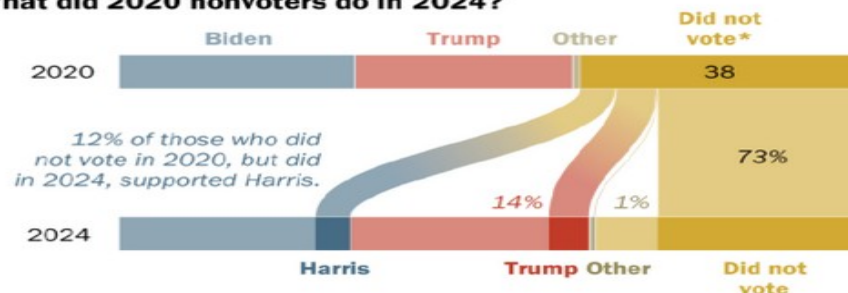
What did Trump's 2020 voters do in 2024?



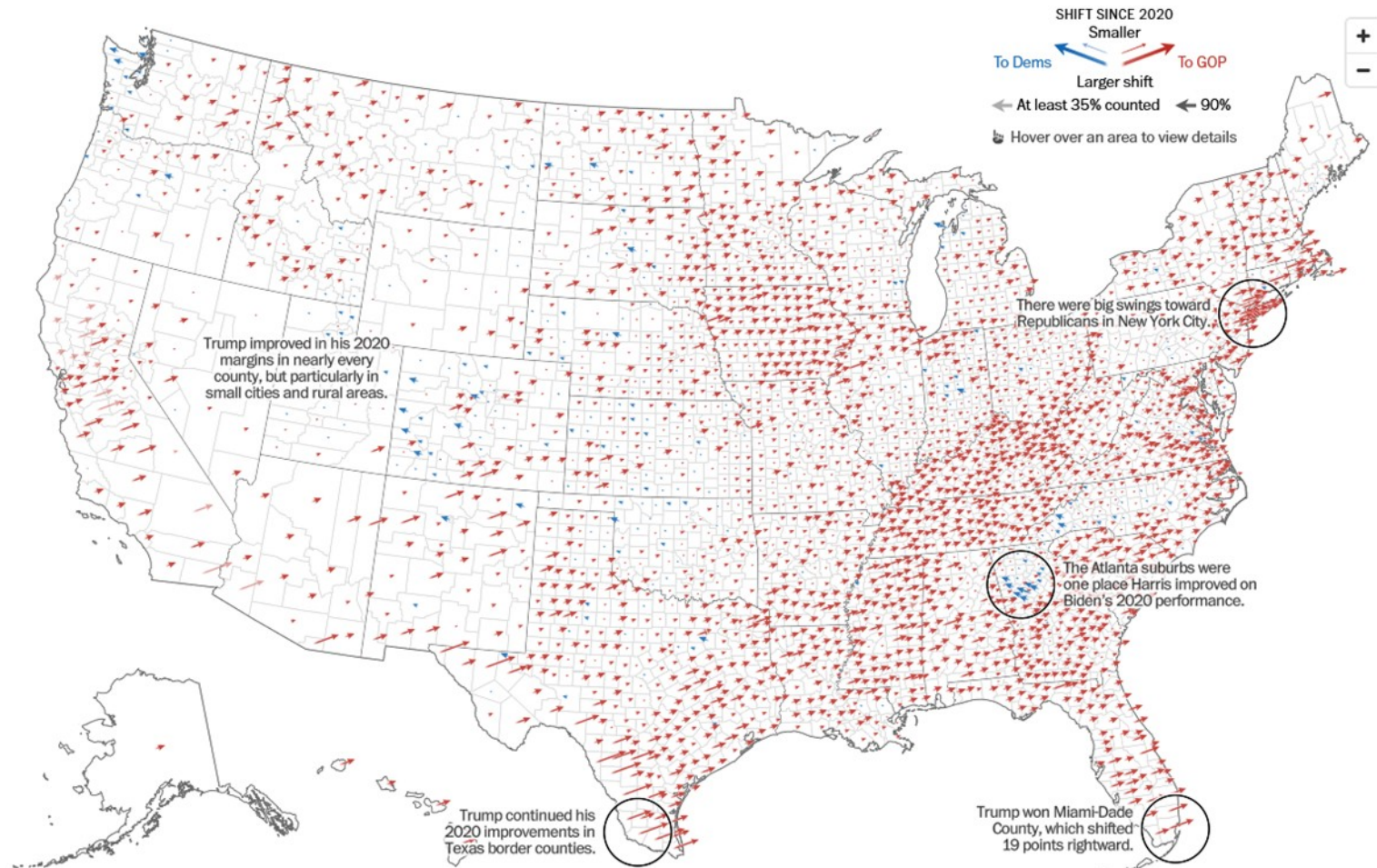
What did Biden's 2020 voters do in 2024?



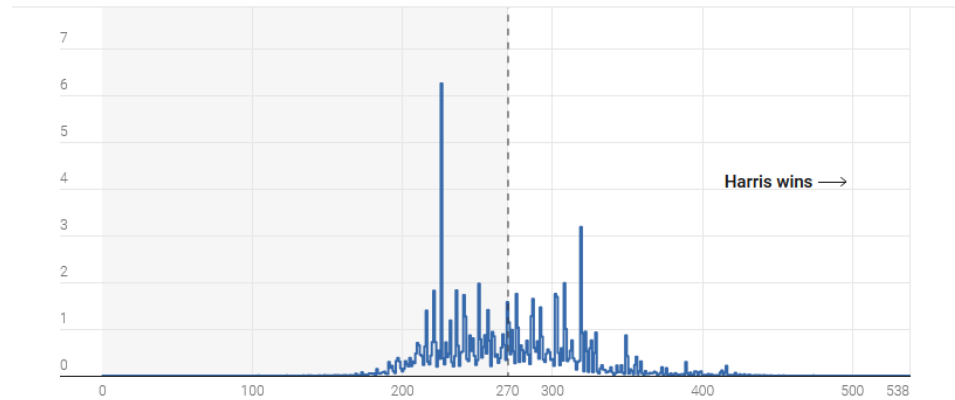
What did 2020 nonvoters do in 2024?



Statistical Consulting for the Media

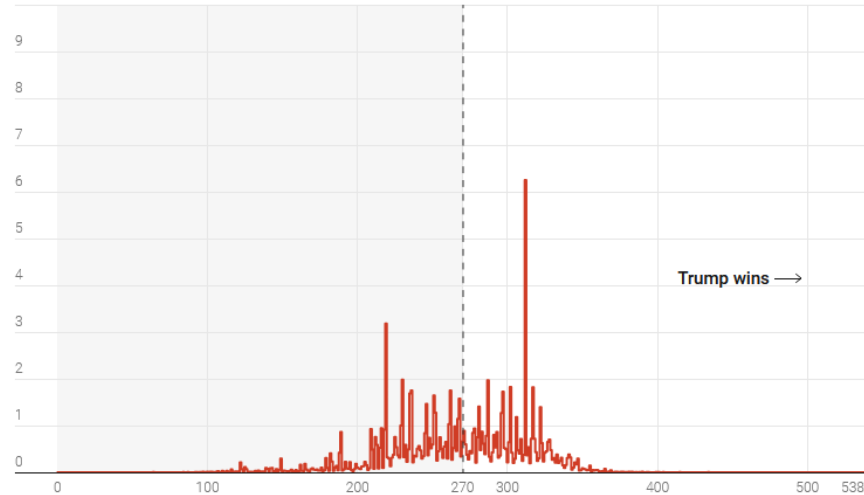


Statistical Consulting for the Media



Trump's electoral votes

10% Chance



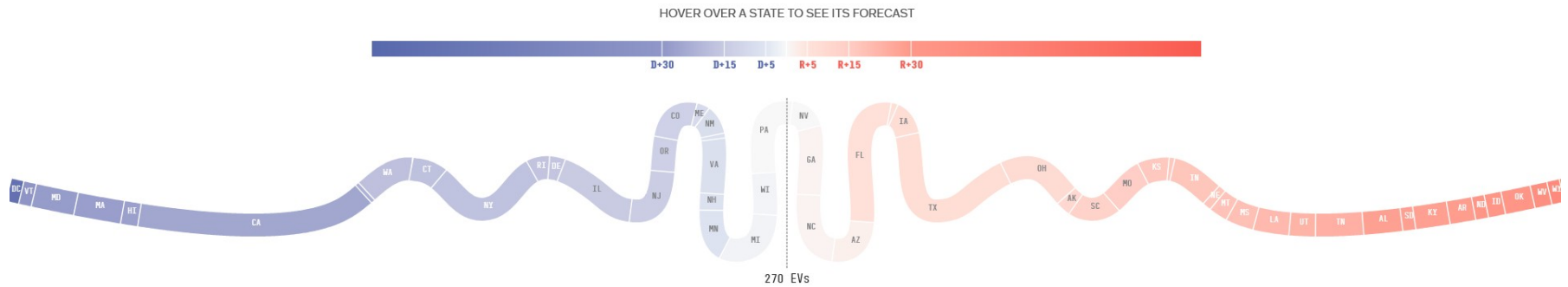
Note: The dotted line indicates 270 electoral votes.

Updated November 5, 2024 - [Get the data](#)

**SILVER
BULLETIN**

Statistical Consulting for the Media

contested states are in the middle. Bigger segments mean more Electoral College votes.



Statistical Consulting for the Media

Work for Gallup in 2016

How to work with a large quantity of new data

Analysis of 500 CATI interviews every night and presentation of results

Use of open end questions with verbatim responses to analyze:

Have you heard, read or seen anything about (Candidate A) in the last couple of days?

IF YES, What was that?

Capture of social media data (Tweets) with a daily sample

Analysis with LLM for topics and sentiment

Statistical Consulting for the Media

Work for Gallup in 2012

GALLUP

June 4, 2013

**GALLUP 2012 PRESIDENTIAL
ELECTION POLLING REVIEW**



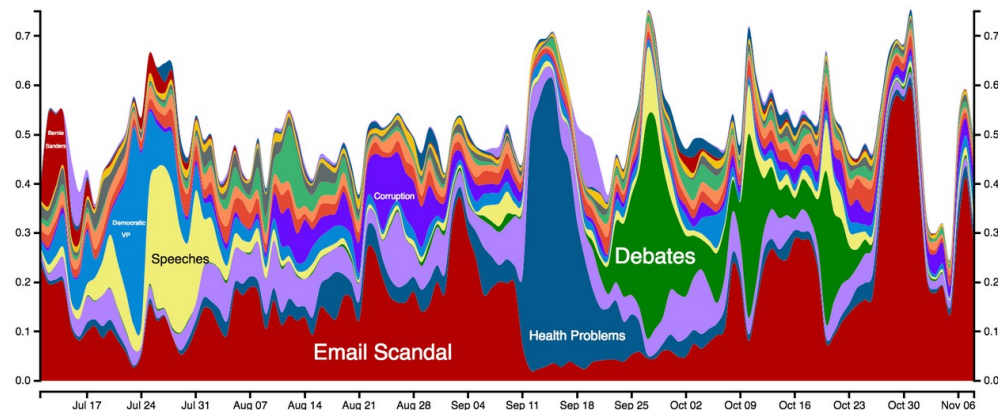
- 4 factors leading to overestimating Romney support
- Likely voter model need for revision
- Regional controls on interviewing
- Race/ethnicity data collection and weighting use
- Ratio of listed landline/cell phone samples

Statistical Consulting for the Media

Work for Gallup in 2016

The political communication project studies how impressions of political candidates are formed by focusing on the role of media representations of them in conjunction with discussions on social media.

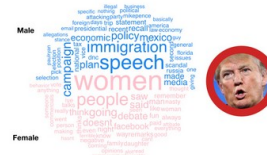
There are two distinct components to this research: an analysis of elite behavior in the press, and an examination of how news coverage and Twitter commentary affect citizens' impressions of the candidates.



This figure highlights the changing topics that Americans remember about Clinton since July. The x-axis shows the date and the y-axis the fraction of responses that fall into a particular topic.

Elite behavior in the media

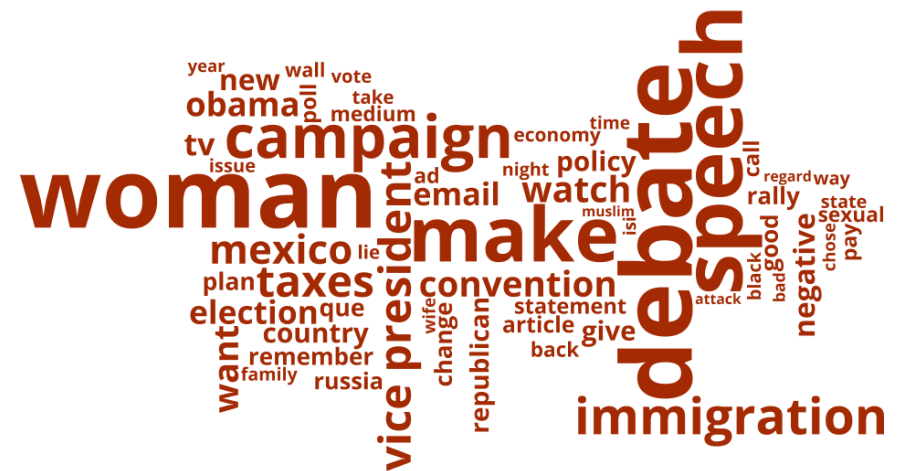
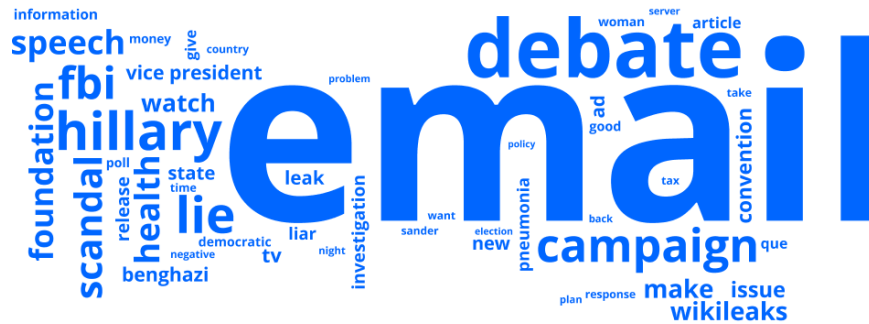
In the analysis of elite behavior, researchers focus on the news content in the mainstream media for two weeks leading up to specific campaign events as well as two weeks after. They examine the construction of the news through journalists' interactions with each other, prior to, during and after important campaign events through Twitter, the platform journalists now see as an essential part of their job.



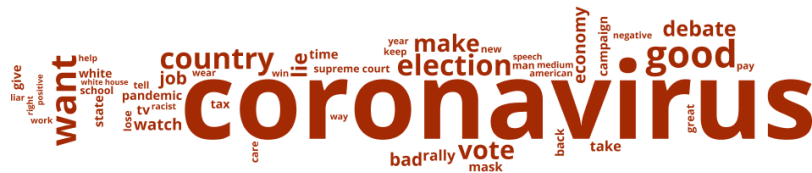
What survey respondents heard about Trump, by sex

Statistical Consulting for the Media

Work for Gallup in 2016



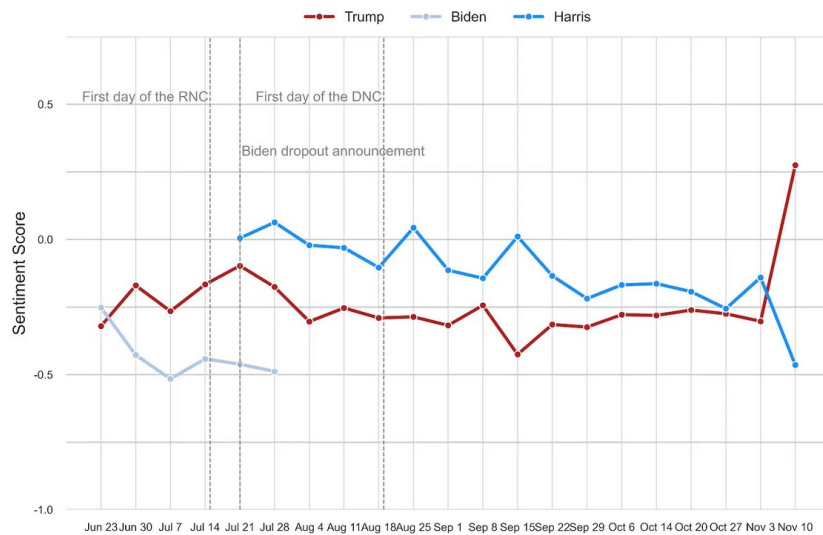
Work for CNN in 2020



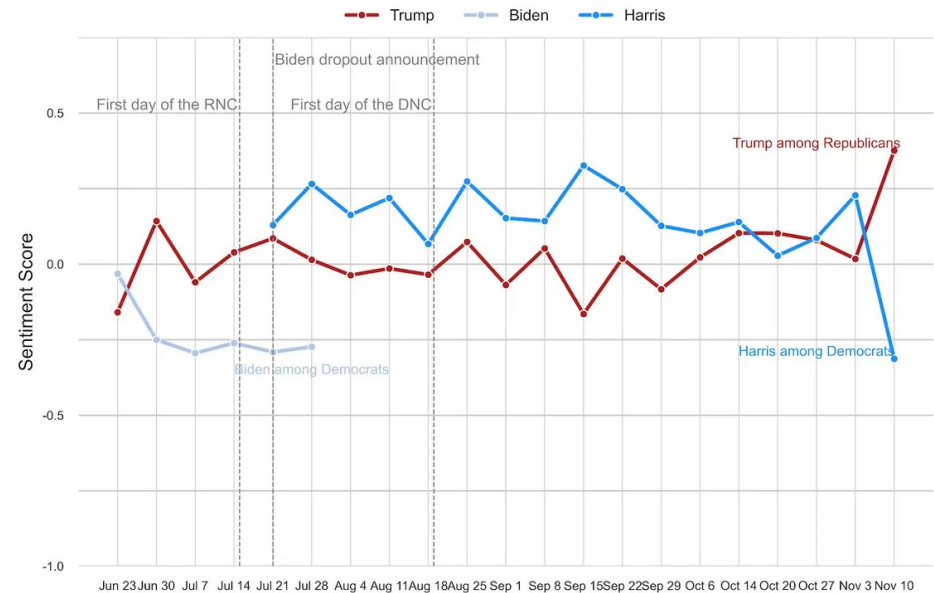
Statistical Consulting for the Media

Work for CNN in 2024

Overall sentiment toward Trump, Biden and Harris



In-Party Sentiment towards Trump, Biden and Harris



Statistical Consulting for the Media

Work for CNN in 2024

≡ CNN Politics SCOTUS Congress Facts First 2024 Elections

⌕ Watch 🎧 Listen

We asked Americans what they'd heard about Trump and Harris throughout the campaign. Here's what they told us



By Ariel Edwards-Levy, CNN

🕒 5 minute read · Published 12:30 PM EST, Sun November 10, 2024

Words by week - Trump

Jun 21–24 felon convicted debate guilty campaign found charges rally court polls race lie view
Jun 28–Jul 1 **debate** presidential watched winning felony convicted race
Jul 5–8 debate court lie supreme felony immunity serving presidential convicted election polls
Jul 12–15 shot attempt assassinate rally project 2025 lie debate war treaty
Jul 19–22 assassinate attempt shot rnc ear speech
Jul 26–29 shot assassinate attempt rally ear debate vp senate campaign wars
Aug 2–5 black assassinate attempt journalist shot interview debate racist campaign no rely
Aug 9–12 debate conference press campaign lie vp assassinate attempt war
Aug 16–19 campaign lie interview usa assassinate election rely election
Aug 23–26 kennedy campaign lie election pro rely usa
Aug 30–Sep 3 arlington cemetery campaign election debate national wars
Sep 6–9 debate sentence election conviction attempt
Sep 13–16 **debate** eating pet lie dog immigrants cat watched who wrong report wars
Sep 20–23 attempt assassinate debate lie eating habitat cat golf may lie immigrants
Sep 27–30 assassinate attempt lie news
Oct 4–7 hurricane lie rally election assassination attempt campaign wars
Oct 11–15 lie rally campaign hurricane recovery mental lie
Oct 18–21 rally lie dinner polls interview ad election wars campaign
Oct 25–28 mcdonalds wars rally usa lie house initial campaign
Nov 1–4 garbage rally truck wants campaign called lie formula election usa mcdonalds mass country polls mental ad

Words by week - Harris

Jul 19–22 democratic nominee endorse presidential candidate democrats drop replace no campaign
Jul 26–29 democratic nominee candidate democrats campaign endorse presidential raise obamasen party no
Aug 2–5 democratic vp candidate nominee presidential raise campaign black usa
Aug 9–12 **vp** chose walz campaign rally debate tim polls democratic candidate
Aug 16–19 campaign vp polls policies economy plan rely democratic price
Aug 23–26 dnc democratic nomination speech convention accepted usa watched candidate presidential party
Aug 30–Sep 3 **interview** cnn campaign vp race polls policies
Sep 6–9 debate campaign polls vp ad policies
Sep 13–16 **debate** winning lie watched presidential good
Sep 20–23 debate interview polls campaign lie question
Sep 27–30 border polls campaign lie ad debate interview wars plan
Oct 4–7 campaign hurricane ad senate war good
Oct 11–15 interview campaign ad hurricane help polls news
Oct 18–21 **interview** fox ad campaign rely good
Oct 25–28 campaign rally interview polls lie lie wars ad class
Nov 1–4 campaign rally ad wants make lie status polls class formula election media vote usa

Statistical Consulting for Private Clients

a. XXX is performing nearly xx% of dx tests in the U.S. How will we show that our sample is representative or skewed (and in what ways) from the entire U.S. tested population?

This is a difficult problem in the absence of convenient nationwide data about the US tested population. Individual states are releasing data, however, so we could pick a subsample of key states – say those in which you have high degrees of market penetration and those in which you have low market penetration, and see the extent to which you are similar or different with respect to age, sex, and/or race (these are the variables generally being reported). A more substantial effort would involve compiling “by hand” the nationwide data (or near nationwide – 4 states are not releasing this data yet) and comparing.

b. Do we include PCR-neg patients in addition to the positives, or only the positives? There is a constraint on the N in our registry and some don’t want to “waste” resources on the negatives. I think it could be valuable in the following way. Tests were and are being done on patients who present with non-specific symptoms suggestive of COVID-19. Comparing positives vs negatives would give insight into the differential diagnosis as we would see what diagnoses resulted for all the people with negative PCR tests. It would also be valuable to compare hospitalization rates and fatality rates between the positives and negatives to get an idea of how much “excess mortality” is associated with COVID-19. So, 2 decisions here – do we include any PCR-negative patients, and if so, in what ratio of positives-to-negatives, bearing in mind that statistical power will be needed for queries about susceptibility to SARS-CoV-2 infection, but not severity of disease.

I would be strongly in favor of collecting data on both positive and negative cases, since in the absence of the latter you are left guessing as to whether what you are seeing in your positive patients. To give an idea of tradeoffs, I’ll provide power calculations under different ratios of positive to negatives: 1:1, 3:1, 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, and 19:1 (i.e., 50% positive, 75% positives, 90% positives, and 95% positives) for overall sample sizes of 15,000, 20,000, 30,000 and 40,000. (Power, just as a reminder, is the probability of detecting a statistically significant difference in a given sample for a true difference of a given size, at a specific level of the hypothesis test. Standards for power are usually 80% probability of detection [i.e. Type II error of 20%] for a Type I error of 5%.) I will show the results for a [comparisons of proportions](#) among the two groups (where one proportion’s true value is 0.5) as well as for comparisons of continuous variables using effect sizes (differences in means divided by a common standard deviation)

This memo is going to be a bit free-flowing, as I try to understand as best as I can the issues involving the sample design for the XXX Market Share Survey.

Although the unit of analysis is the package, as a practical matter you will have to sample “shippers.” I use that term in quotes because this is a somewhat ill-defined population that consists of all a) persons who sent small packages (defined as 0-150 lbs) during the past year, and b) of all organizations that sent small packages during the past year, subdivided down to the unit at which records are tracked as to the details of the small package shipping decisions, including the choice of the vendor that sends a given small package. Persons, of course, can appear in both populations, either representing themselves in the personal life, if you will, or their organizations in their work, professional, or personal life.

It is relatively easy to deal with the former; given the ubiquity of small package shipment, I might simply suggest a random-digit-dial survey. You might want to consider asking about the past month, however, given that personal shipments might be considerably less common than organizational shipments.

For the latter, the trick will be obtaining a sampling frame that corresponds with the population. Marrying the conceptual and practical here is difficult. As a start I imagine that, in principle at least, you can obtain a sampling frame of businesses stratified by SIC or NAICS code and size. You sample from within these strata at random with a known probability of selection. The probability of selection can differ across strata or even within strata (for example you might want to select a few major retailers, particularly mail order retailers, with probability 1), as long as you maintain a record of what the probability of selection is. Once you have a business, you may need to have a second or even third round of sampling within the facility (this will probably have to be at random) if there is no centralized recording of small package shipments within the business. For example, if you sample LL Bean, it may be that they can tell you to the nearest dollar the nine-figure amounts of shipping costs by shipper, but at the University of Pennsylvania, it is likely that these data are kept, at best, at a department or division level; that would require getting a list of departments, sampling n out of N of them at random, and retaining this second-stage sampling fraction record for construction of future weights.

To construct the sampling weights for the i th shipper in j th business in the h th stratum, let n_h be the number of business sampled from the h th stratum at random from N_h businesses, and m_{jh} be the number of shippers in the j th business, drawn at random from the M_{jh} shippers determined to exist in the j th business. Then the sampling design weight would be

$$d_{ijh} = \frac{M_{jh} N_h}{m_{jh} n_h}.$$

If there are known control totals X_h (e.g., total sales, etc.) within a given stratum, one can construct calibrated or “[poststratified](#)” weights w_{ijh} such that $\sum_i \sum_j w_{ijh} x_{ijh} = X_h$ using

generalized regression or generalized raking estimators. UPS may have some specific control totals of interest at the SIC-level to which they may want your weight totals to correspond.

Statistical Consulting for Academic Clients

Rather than conduct a pure power analysis, I have iterated between from your \$700,000 budget for data collection and the power available under a variety of affordable sample sizes. I am proposing a standard two-stage area probability sample, with 30 blocks groups to be sampled with probability proportional to size, and a fixed number of households to be sampled at random within each of the sampled block groups and successfully screened and recruited into the study (assuming one African-American adult 18-55 is further selected at random from each eligible and participating household). The use of 30 blocks (of the 190 across the two neighborhoods) allows for a reasonably stable estimate of between- and within-neighborhood variance to be obtained. Assuming it will cost ~\$1,000 to successfully screen, recruit, and obtain a single wave of data collection and several hundred dollars for each additional wave, this suggests between 10 and 15 households be sampled within each sampled block group, yielding a total sample size of between 300 and 450. (Costs for such studies are highly variable, and I am not a project manager and I am not guaranteeing the dollar values, but I think this should get us in the ballpark.)

Aim 1 analysis

For the Aim 1 analysis, I will consider the power to detect linear relationships between inflammatory markers (e.g., sIL-R6) and measures of individual-level racism experience. To simplify calculations and interpretation, I will assume that the measures of individual-level racism experience x_{it} for subject i and time t have been standardized to have mean 0 and variance 1, and that the measures of inflammatory markers y_{it} have also been standardized to have mean 0 and variance 1, after any appropriate transformations (e.g., log) have been made to approximate normality. Thus slopes can be interpreted as the fraction of a standard deviation (SD) change in inflammatory marker for each SD change in of individual-level racism. We then assume a model is being fit of the form:

$$y_{it} = \beta_0 + \beta_1 x_{it} + \sum_{k=2}^K \beta_k z_{it} + b_{i0} + b_{i1} x_{it} + \varepsilon_{it}$$

$$b_{i0} \sim N(0, \tau_0) \quad b_{i1} \sim N(0, \tau_1) \quad \varepsilon_{it} \sim N(0, \sigma_y^2)$$

that is, a random-effect slope-intercept model with subject-level slopes and intercepts $\beta_0 + b_{i0}$ and slopes $\beta_1 + b_{i1}$ associated with individual level affects of racism experience after adjusting for fixed covariates z_{it} (time-invariant baseline demographic and neighborhood characteristics in the regression model, but possibly including time-varying mediators of interest as well). We assume that the fraction of variance explained by the fixed z_{it} (r^2) is .1 or .4. We divide up the between- and within- subject variance under the assumption of little within-subject correlation and large within-subject correlation, and assume complete follow-up after recruitment and initial interview. Adjusting the

effective sample size to account for estimated design effects of 1.2 due to weighting (the household selection will be equal probability, but selection weights will be required to account for the underrepresentation of subjects in household with multiple African-Americans 18-55, and additional weights may be required to account for non-response and poststratification), and also allowing for possible residual clustering effects by neighborhood yields the following detectable slopes β_1 with 80% power under a Type I error of $\alpha = .05$:

N=300, $r^2=.1$			ρ (within-neighborhood correlation of inflammatory marker)		
σ_y^2	τ_0^2	τ_1^2	0	.02	.05
.846	.045	.009	.074	.081	.091
.45	.27	.18	.085	.094	.105

N=300, $r^2=.4$			ρ (within-neighborhood correlation of inflammatory marker)		
σ_y^2	τ_0^2	τ_1^2	0	.02	.05
.546	.030	.006	.059	.065	.073
.30	.18	.12	.069	.076	.085

N=450, $r^2=.1$			ρ (within-neighborhood correlation of inflammatory marker)		
σ_y^2	τ_0^2	τ_1^2	0	.02	.05
.846	.045	.009	.074	.081	.091
.45	.27	.18	.085	.094	.105

N=450, $r^2=.4$			ρ (within-neighborhood correlation of inflammatory marker)		
σ_y^2	τ_0^2	τ_1^2	0	.02	.05
.546	.030	.006	.059	.065	.073
.30	.18	.12	.057	.065	.075